

Surveillance System

Project Aim

Design and implement a distributed surveillance system for threat detection, alert generation, and barrier control.

Project Overview

The Smart Room Entry System is an Arduino-based automation project that uses two Arduino Uno boards communicating through serial communication. An ultrasonic sensor detects visitors, automatically opens and closes a door using a servo motor, controls room lighting, monitors occupancy, and activates a fan when a visitor remains present for an extended period.

Components Used

- Arduino Uno (3 Nos.)
- PIR Motion Sensor
- TMP36 Temperature Sensor
- LDR (Light Dependent Resistor)
- 10k Ω Resistor (for LDR voltage divider)
- 16 $\times$ 2 LCD Display (2 Nos.)
- 10k Ω Potentiometer (2 Nos., one for each LCD)
- Servo Motor (Security Barrier)
- Green LED
- Yellow LED
- Red LED
- Piezo Buzzer
- Breadboards
- Jumper Wires

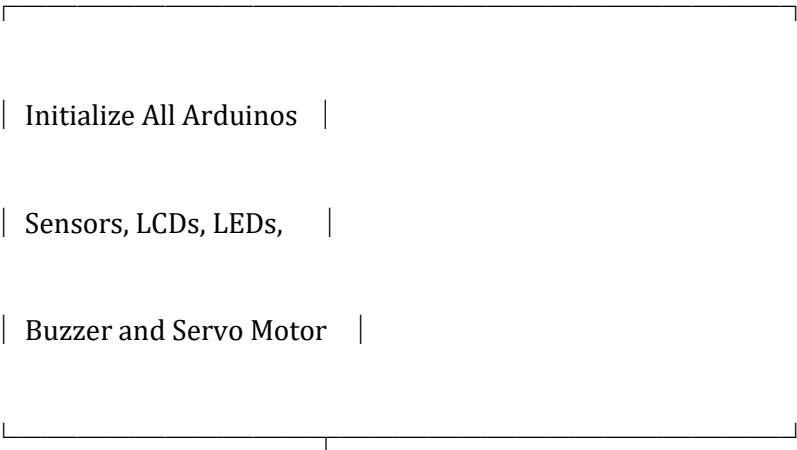
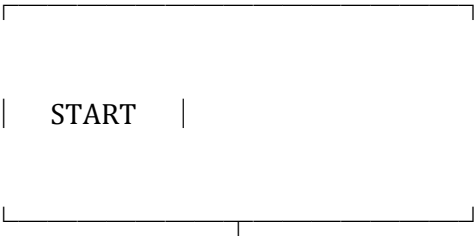
Construction and Wiring Details

Arduino	Component	Component Pin	Arduino Pin
Slave 1	PIR Sensor	VCC	5V
Slave 1	PIR Sensor	GND	GND
Slave 1	PIR Sensor	OUT	D2
Slave 1	TMP36	Left Pin	5V
Slave 1	TMP36	Middle Pin	A0
Slave 1	TMP36	Right Pin	GND
Slave 1	LDR	Terminal 1	5V
Slave 1	LDR	Terminal 2	A1
Slave 1	10k Ω Resistor	End 1	A1
Slave 1	10k Ω Resistor	End 2	GND
Slave 1	UART TX	D1	Master D0
Master	LCD 1	RS	D2
Master	LCD 1	E	D3
Master	LCD 1	D4	D4
Master	LCD 1	D5	D5
Master	LCD 1	D6	D6
Master	LCD 1	D7	D7
Master	LCD 1	VSS	GND
Master	LCD 1	VDD	5V

Arduino	Component	Component Pin	Arduino Pin
Master	LCD 1	RW	GND
Master	LCD 1	V0	Potentiometer 1 Middle Pin
Master	LCD 1	A (LED+)	5V
Master	LCD 1	K (LED-)	GND
Master	Potentiometer 1	Left Pin	GND
Master	Potentiometer 1	Middle Pin	LCD 1 V0
Master	Potentiometer 1	Right Pin	5V
Master	LCD 2	RS	A2
Master	LCD 2	E	A3
Master	LCD 2	D4	A4
Master	LCD 2	D5	A5
Master	LCD 2	D6	D8
Master	LCD 2	D7	D9
Master	LCD 2	VSS	GND
Master	LCD 2	VDD	5V
Master	LCD 2	RW	GND
Master	LCD 2	V0	Potentiometer 2 Middle Pin
Master	LCD 2	A (LED+)	5V
Master	LCD 2	K (LED-)	GND
Master	Potentiometer 2	Left Pin	GND
Master	Potentiometer 2	Middle Pin	LCD 2 V0
Master	Potentiometer 2	Right Pin	5V

Arduino	Component	Component Pin	Arduino Pin
Master	UART TX	D1	Slave 2 D0
Slave 2	Green LED	Anode (+)	D2 through 220Ω
Slave 2	Green LED	Cathode (-)	GND
Slave 2	Yellow LED	Anode (+)	D3 through 220Ω
Slave 2	Yellow LED	Cathode (-)	GND
Slave 2	Red LED	Anode (+)	D4 through 220Ω
Slave 2	Red LED	Cathode (-)	GND
Slave 2	Buzzer	Positive (+)	D5
Slave 2	Buzzer	Negative (-)	GND
Slave 2	Servo Motor	Signal	D6
Slave 2	Servo Motor	VCC	5V
Slave 2	Servo Motor	GND	GND
All Arduinos Common Ground GND			Connected Together

Code Structure / Program Flow



| · Temperature Sensor |

| · LDR Sensor |

|

|



|

| Send Sensor Data to |

| Master via UART |

|

|



|

| Master Receives Data |

| and Displays Temp & |

| Light on LCD 1 |

|

|



|

| Analyze Threat Level |

|

|



|

| Motion = 0 ? |

|

Yes | No

|



|

| Temp $\leq$ 35 AND |

| Light $\geq$ 300 ? |

Yes | | No



| LOW Threat | | MEDIUM |

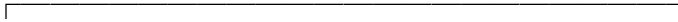
| Green LED | | Threat |

| ON | | Yellow LED |



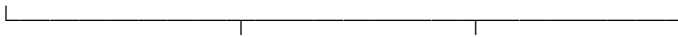
| Motion = 1 ? |

|



| Temp $\leq$ 35 AND |

| Light $\geq$ 300 ? |



Yes | | No



| HIGH | | CRITICAL |

| Threat | | Threat |

| Red LED ON | | Red LED ON |

| Buzzer ON | | Buzzer ON |

| Barrier | | Barrier |

| Closed | | Open 90° |



| |

|-----|

|



|-----|

| Display Threat & Barrier |

| Status on LCD 2 |

|-----|

|



|-----|

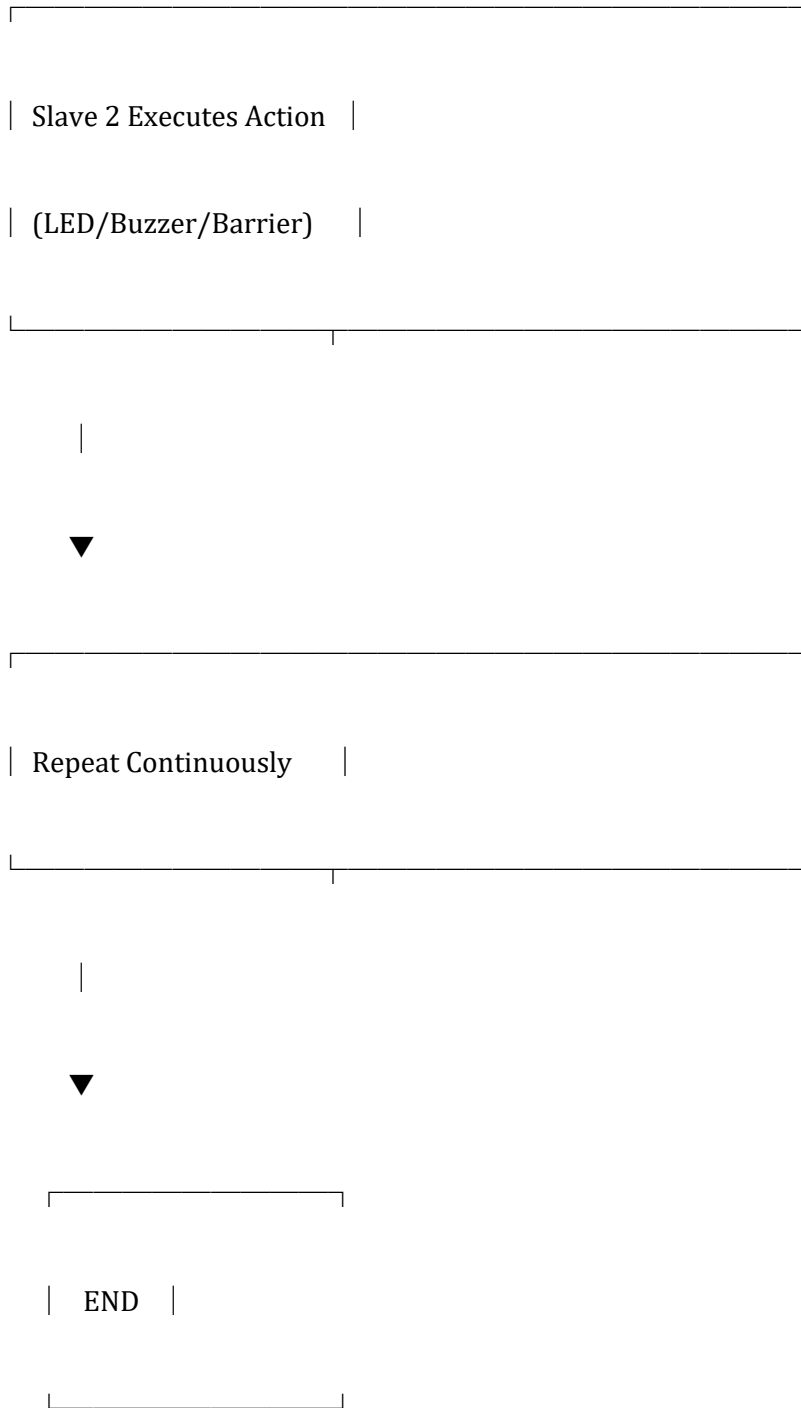
| Send Command to Slave 2 |

| via UART |

|-----|

|





Source Code

Arduino 1 Slave 1

```
const int pirPin = 2;
```

```
const int tempPin = A0;
const int ldrPin = A1;

void setup()
{
  Serial.begin(9600);
  pinMode(pirPin, INPUT);
}

void loop()
{
  int motion = digitalRead(pirPin);

  long total = 0;

  for(int i=0;i<20;i++)
  {
    total += analogRead(tempPin);
    delay(5);
  }

  float raw = total / 20.0;
  float voltage = raw * 5.0 / 1023.0;
  float temp = (voltage - 0.5) * 100.0;

  int light = analogRead(ldrPin);

  Serial.print(motion);
```

```
Serial.print(",");  
Serial.print((int)temp);  
Serial.print(",");  
Serial.println(light);  
  
delay(2000);  
}
```

Master Arduino 2

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd1(2,3,4,5,6,7);
```

```
LiquidCrystal lcd2(A2,A3,A4,A5,8,9);
```

```
String data="";
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  lcd1.begin(16,2);
```

```
  lcd2.begin(16,2);
```

```
  lcd1.print("SYSTEM READY");
```

```
  lcd2.print("WAITING");
```

```
  delay(2000);
```

```
lcd1.clear();
```

```
lcd2.clear();
```

```
}
```

```
void loop()
```

```
{
```

```
if(Serial.available())
```

```
{
```

```
data = Serial.readStringUntil('\n');
```

```
int c1 = data.indexOf(',');
```

```
int c2 = data.lastIndexOf(',');
```

```
if(c1==-1 || c2==-1) return;
```

```
int motion =
```

```
data.substring(0,c1).toInt();
```

```
int temp =
```

```
data.substring(c1+1,c2).toInt();
```

```
int light =
```

```
data.substring(c2+1).toInt();
```

```
lcd1.clear();
```

```
lcd1.setCursor(0,0);
```

```
lcd1.print("TEMP:");
```

```
lcd1.print(temp);
```

```
lcd1.print("C");
```

```
lcd1.setCursor(0,1);
```

```
lcd1.print("LIGHT:");
```

```
lcd1.print(light);
```

```
String threat;
```

```
String barrier;
```

```
if(motion==0 && temp<=35 && light>=300)
```

```
{
```

```
    threat="LOW";
```

```
    barrier="CLOSED";
```

```
    Serial.println("LOW");
```

```
}
```

```
else if(motion==0 && (temp>35 || light<300))
```

```
{
```

```
    threat="MEDIUM";
```

```
    barrier="CLOSED";
```

```
    Serial.println("MEDIUM");
```

```
}
```

```
else if(motion==1 && temp<=35 && light>=300)
```

```
{
```

```
    threat="HIGH";
```

```
    barrier="CLOSED";
```

```
    Serial.println("HIGH");  
}  
  
else  
{  
    threat="CRITICAL";  
    barrier="OPEN";  
    Serial.println("CRITICAL");  
}
```

```
lcd2.clear();
```

```
lcd2.setCursor(0,0);  
lcd2.print("THREAT:");  
lcd2.print(threat);
```

```
lcd2.setCursor(0,1);  
lcd2.print("BARRIER:");  
lcd2.print(barrier);  
}  
}
```

Slave 2 Arduino 3

```
#include <Servo.h>
```

```
Servo barrier;
```

```
const int greenLED = 2;  
const int yellowLED = 3;
```



```
const int redLED = 4;
const int buzzer = 5;
const int servoPin = 6;
```

```
String command="";
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  pinMode(greenLED, OUTPUT);
```

```
  pinMode(yellowLED, OUTPUT);
```

```
  pinMode(redLED, OUTPUT);
```

```
  pinMode(buzzer, OUTPUT);
```

```
  barrier.attach(servoPin);
```

```
  barrier.write(0);
```

```
}
```

```
void resetOutputs()
```

```
{
```

```
  digitalWrite(greenLED, LOW);
```

```
  digitalWrite(yellowLED, LOW);
```

```
  digitalWrite(redLED, LOW);
```

```
  digitalWrite(buzzer, LOW);
```

```
  barrier.write(0);
```

```
}
```

```
void loop()
{
  if(Serial.available())
  {
    command = Serial.readStringUntil('\n');
    command.trim();

    resetOutputs();

    if(command=="LOW")
    {
      digitalWrite(greenLED,HIGH);
    }

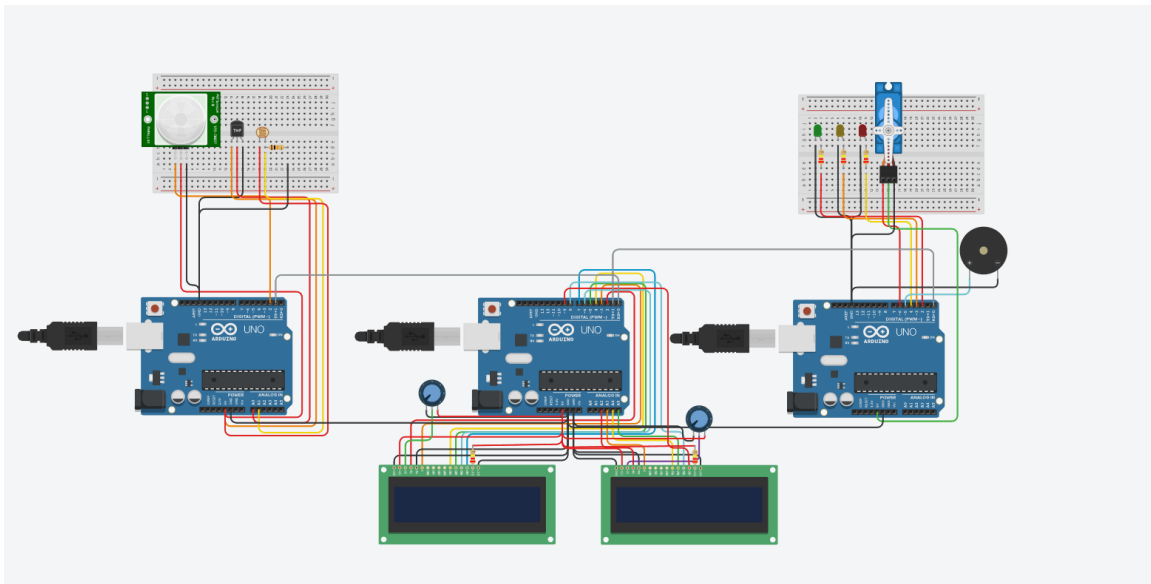
    else if(command=="MEDIUM")
    {
      digitalWrite(yellowLED,HIGH);
    }

    else if(command=="HIGH")
    {
      digitalWrite(redLED,HIGH);
      digitalWrite(buzzer,HIGH);
    }

    else if(command=="CRITICAL")
    {
```

```
digitalWrite(redLED,HIGH);  
  
digitalWrite(buzzer,HIGH);  
  
barrier.write(90);  
  
}  
  
}  
  
}
```

Project Photograph



Links

Google drive : <https://drive.google.com/drive/folders/1gE1CIIHHW-E21ryxjDBqySG1C78Y5ZSQ?usp=sharing>

Tinkercad :
<https://www.tinkercad.com/things/3DuYHrBxOpc/editel?returnTo=%2Fdashboard&sharecode=C6SUq2ULFMoWm2O-M3t2BOPR9XFWu91xD5OU2Wzl2Wg>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>